CUHKFAA Chan Chun Ha Secondary School

Final Examination (2024-2025)

Secondary Two MATHEMATICS

PAPER 2

10:05a.m. -10:50a.m.

2-6-2025

Time allowed: 45 minutes

Name:_____(_____) Class:_____ Marks:_____/30

1. Read the instructions carefully on the Question Paper.

2. When told to read this paper, you should check that the question paper has no missing pages. Look for the words '**END OF PAPER**' on the last page.

3. All questions carry equal marks.

4. **ANSWER ALL QUESTIONS.** You are advised to mark all the answers on the Multiple-choice Answer Sheet with an HB pencil, so that wrong marks can be completely erased with a clean rubber.

5. You should mark only **ONE** answer to each question. Two or more answers will score **NO MARKS**.

6. No marks will be deducted for wrong answers.

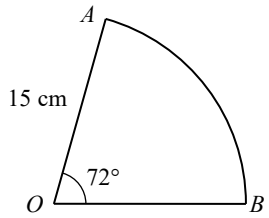
There are 20 questions in Section A and 10 questions in Section B.

The diagrams in this paper are not necessarily drawn to scale.

Choose the best answer for each question.

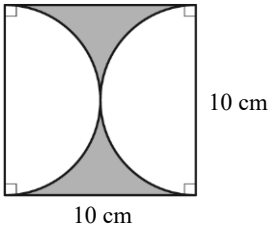
Section A

1. In the figure, the radius of sector OAB is 15 cm and $\angle AOB = 72°$. Find the arc length of the sector, correct to the nearest 0.01 cm.



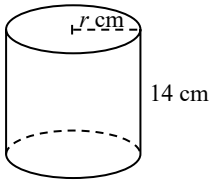
A. 9.42 cm
B. 18.85 cm
C. 67.86 cm
D. 141.37 cm

2. Find the area of the shaded region in the following figure. (Take $\pi = 3.14$.)



A. 21.5 cm^2
B. 46.5 cm^2
C. 53.5 cm^2
D. 214 cm^2

3. The figure shows a right cylinder. Its height is 14 cm and its volume is 504π cm^3. The base radius of the cylinder is r cm. Find the value of r.

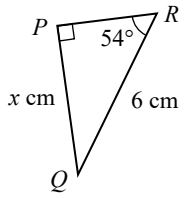


A. 6
B. 9
C. 12
D. 18

4. If $\cos\theta = 0.483$, then $\theta =$

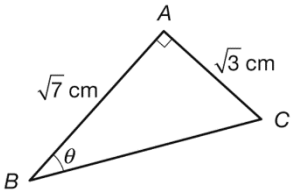
A. 25.8°. *(cor. to 3 sig. fig.)*
B. 28.9°. *(cor. to 3 sig. fig.)*
C. 48.3°. *(cor. to 3 sig. fig.)*
D. 61.1°. *(cor. to 3 sig. fig.)*

5. In the figure, $x =$



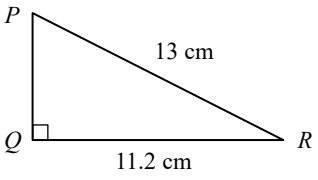
A. 3.53. *(cor. to 3 sig. fig.)*
B. 4.85. *(cor. to 3 sig. fig.)*
C. 7.42. *(cor. to 3 sig. fig.)*
D. 8.26. *(cor. to 3 sig. fig.)*

6. In the figure, $\theta =$



A. 23.2°. *(cor. to 3 sig. fig.)*
B. 33.2°. *(cor. to 3 sig. fig.)*
C. 56.8°. *(cor. to 3 sig. fig.)*
D. 66.8°. *(cor. to 3 sig. fig.)*

7. In the figure, $\angle PQR = 90°$, $PR = 13$ cm and $QR = 11.2$ cm. $PQ =$



A. 1.8 cm.
B. 3.3 cm.
C. 3.6 cm.
D. 6.6 cm.

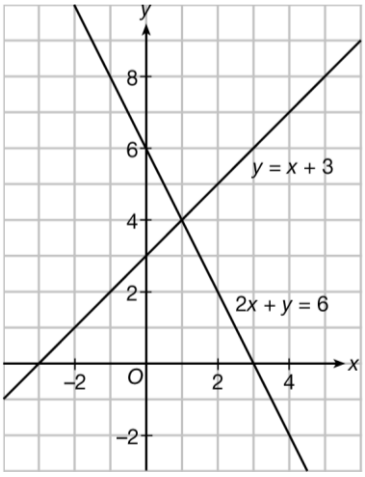
8. In ΔABC, if $AB^2 + AC^2 = BC^2$, then

A. $\angle A + \angle B = 90°$.
B. $\angle B + \angle C = 90°$.
C. $\angle A + \angle C = 90°$.
D. ΔABC is not a right-angled triangle.

9. $\sqrt{20} \times 3\sqrt{3} - 7\sqrt{15} =$

A. $5\sqrt{15}$
B. $3\sqrt{15}$
C. $\sqrt{15}$
D. $-\sqrt{15}$

10. According to the given graphs, solve the simultaneous equations $\begin{cases} y = x + 3 \\ 2x + y = 6 \end{cases}$ graphically.



A. (0.5, 4)
B. (1, 4)
C. (4, 0.5)
D. (4, 1)

11. If $\begin{cases} 7x - 3y = 3 \\ 7x + 2y = 33 \end{cases}$, then $x =$

A. 3.
B. 6.
C. 11.
D. 21.

12. If $x \geq y$, which of the following inequalities is correct?

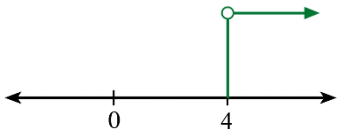
A. $\frac{x}{-9} \geq \frac{y}{-9}$

B. $x - 9 \leq y - 9$

C. $\frac{x}{9} \geq \frac{y}{9}$

D. $9x \leq 9y$

13. The diagram represents the solutions of an inequality.



The least integer satisfying the inequality is

A. 1.

B. 3.

C. 4.

D. 5.

14. The solution of $4 - \frac{x}{3} \geq 3$ is

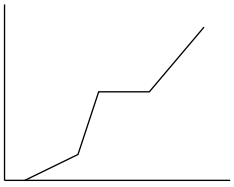
A. $x \leq -5$.

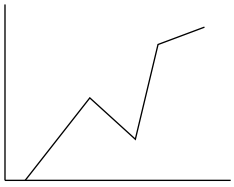
B. $x \geq 3$.

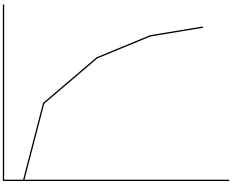
C. $x \leq 3$.

D. $x \geq 21$.

15. Which of the following may be cumulative frequency polygons?

I. 

II. 

III. 

A. I and II only

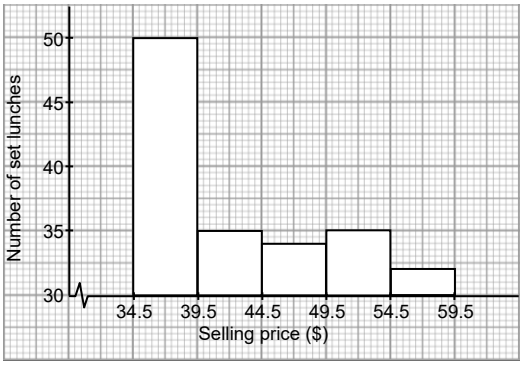
B. I and III only

C. II and III only

D. I, II and III

16. The figure shows the selling prices (in $) of set lunches in a restaurant on one day.

Selling prices of set lunches in a restaurant



Based on the figure above, Mary believes that the selling prices of more than half of the set lunches are between $34.5 and $39.5 on that day.
Which of the following statements is the best reason that Mary is misled by the above figure?

A. The horizontal axis is not marked with the class mark of each class interval.
B. The scale of the vertical axis does not start from 0.
C. The date for collecting the data is not indicated.
D. The selling prices of the set lunches in other restaurants are not compared.

17. The length of a table is 460 cm (correct to the nearest cm). Which of the following is the lower limit and the upper limit of the actual length of the table?

	Lower limit	Upper limit
A.	455 cm	465 cm
B.	459 cm	461 cm
C.	459.45 cm	460.55 cm
D.	459.5 cm	460.5 cm

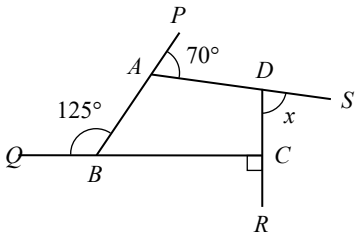
18. $3hx + 4x - 3h - 4 =$

A. $(3h - 4)(x - 1)$
B. $(3h - 4)(x + 1)$
C. $(3h + 4)(x - 1)$
D. $(3h + 4)(x + 1)$

19. $-2(5r + 4)(5r - 4) =$

A. $-50r^2 + 32$
B. $-50r^2 + 80r - 32$
C. $-100r^2 + 64$
D. $-100r^2 + 160r - 64$

20. The figure shows the exterior angles of a quadrilateral $ABCD$. $x =$

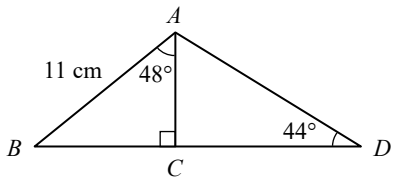


A. 65°.
B. 70°.
C. 75°.
D. 80°.

Section B

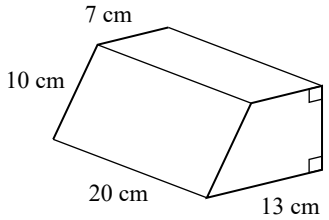
21. The total surface area and the base radius of a cylinder are 130π cm^2 and 5 cm respectively. The volume of the cylinder is
    A. 80π cm^3.
    B. 200π cm^3.
    C. 262.5π cm^3.
    D. 325π cm^3.

22. In the figure, BCD is a straight line. $CD =$



A. 7.11 cm. *(cor. to 3 sig. fig.)*
B. 7.62 cm. *(cor. to 3 sig. fig.)*
C. 7.89 cm. *(cor. to 3 sig. fig.)*
D. 8.47 cm. *(cor. to 3 sig. fig.)*

23. In the figure, the volume of the solid right trapezoidal prism is



A. 1400 cm^3.
B. 1600 cm^3.
C. 1820 cm^3.
D. 2000 cm^3.

24. If $\begin{cases} 5x + 4y = 8 \\ 6x + 5y = 7 \end{cases}$, then $x + y =$

A. -1.

B. 1.

C. 5.

D. 25.

25. The solution of $\frac{3-2x}{7} + \frac{x-7}{4} \leq -1$ is

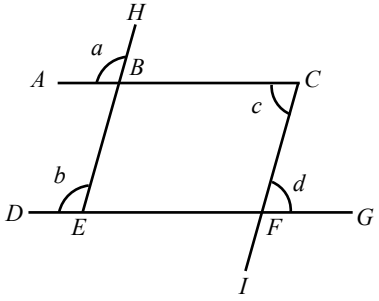
A. $x \geq -9$.

B. $x \leq -9$.

C. $x \geq -36$.

D. $x \leq -36$.

26. In the figure, *ABC*, *DEFG*, *HBE* and *CFI* are straight lines.



If $a = b$, which of the following must be true?

I. $AC // DG$

II. $c = d$

III. $HE // CI$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

27. $\frac{10x^2}{9y^2-16x^2} \div \frac{6x}{16x+12y} =$

A. $\frac{20x}{3(4x-3y)}$

B. $\frac{20x}{3(3y-4x)}$

C. $\frac{5x}{12(4x-3y)}$

D. $\frac{5x}{12(3y-4x)}$

28. If $y = \frac{px+1}{q-x}$, then $x =$

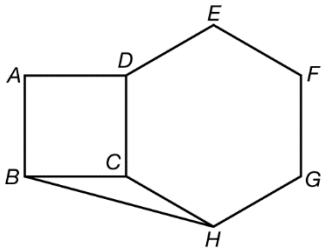
A. $\frac{1-qy}{y+p}$.

B. $\frac{1-q}{1+p}$.

C. $\frac{qy-1}{p+y}$.

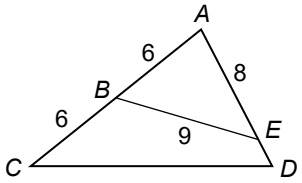
D. $\frac{q-1}{p+1}$.

29. In the figure, *ABCD* is a square and *CDEFGH* is a regular hexagon. Find $\angle CBH$.



A. 12°
B. 15°
C. 20°
D. 25°

30. In the figure, $\Delta ACD \sim \Delta AEB$. Find the perimeter of ΔACD.



A. 33
B. 34.5
C. 36
D. 37.5

END OF PAPER